Full derivative Class-II Gaussian ultraviolet power counting

TECT verification-first repository · claim A6-CLASSII-UV-POWER-COUNTING

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1 Purpose and scope

The scalar constructive theorem does not contain the derivative Class-II energy. This note asks the prior question needed before attempting such a construction: under the Gaussian measure fixed by the quadratic part of the canonical full-production functional, what is the ultraviolet growth of the bare currents and what field-dependent local contraction is generated?

The domain is the fixed three-torus of side $L = 16$. The field has three complex components and is projected to $\max_j |n_j| \le N$, with $k_n = 2\pi n/L$. The positive density and Class-II mass regularisers remain fixed, and $\eta_{\text{shell}} = 0$. This is not a construction of an interacting measure.

2 Quadratic Gaussian convention

The quadratic symbol is

$$A(k) = \big(r + Z|k|^2 + Y|k|^4\big)I_3 + M_{\text{family}} + k_{\text{lock}}(I - P_0).$$

The real and imaginary parts of every complex mode are six real Gaussian coordinates with covariance $A(k)^{-1}$. Therefore the load-bearing complex covariance is

$$\mathbb{E}[\Psi_k \Psi_k^\dagger] = 2A(k)^{-1}.$$

With the volume-normalised Fourier convention, define

$$C_N = \frac{2}{V} \sum_{\|n\|_\infty \le N} A(k_n)^{-1}, \qquad D_N = \frac{2}{3V} \sum_{\|n\|_\infty \le N} |k_n|^2 A(k_n)^{-1}.$$

Cube symmetry makes D_N the derivative covariance in each spatial direction. The odd value–derivative covariance vanishes exactly, so value and derivative are independent.

3 Derivative-covariance theorem

For large momentum,

$$A(k)^{-1} = \frac{1}{Y|k|^4} I_3 + O(|k|^{-6}).$$

The remainder contributes a convergent derivative-covariance sum. The leading term is a cube Riemann sum:

$$\frac{D_N}{N} \longrightarrow \delta_{\text{cube}} I_3, \qquad \delta_{\text{cube}} = \frac{I_{\text{cube}}}{6\pi^2 Y L},$$

where

$$I_{\text{cube}} = \int_{[-1,1]^3} \frac{dx}{|x|^2}.$$

The singularity is integrable in three dimensions. Using $\nabla \cdot (x/|x|^2) = |x|^{-2}$ gives the nonsingular surface representation

$$I_{\text{cube}} = 6 \int_{[-1,1]^2} \frac{dy\, dz}{1 + y^2 + z^2} = 15.348248444887457\ldots.$$

For $L = 16, Y = 1$,

$$\delta_{\text{cube}} = 0.01619898645075695.$$

In contrast, C_N converges because the summand is $O(|n|^{-4})$.

4 Exact conditional current identities

Let T_A be the three embedded Pauli generators and put

$$\rho = \Psi^\dagger \Psi, \quad m_A = \Psi^\dagger T_A \Psi, \quad q_A = \frac{m_A}{\rho + \varepsilon_\rho},$$

$$J_A = \nabla m_A, \qquad K_A = J_A - q_A \nabla \rho.$$

If $w_i = \partial_i \Psi$ and $v_A = (T_A - q_A I)\Psi$, circular Gaussian contraction conditional on Ψ gives

$$\mathbb{E} \sum_i J_{A,i}^2 = 6\, \mathbb{E}[(T_A \Psi)^\dagger D_N (T_A \Psi)],$$

$$\mathbb{E} \sum_i J_{A,i} K_{A,i} = 6\, \mathbb{E}\, \mathrm{Re}[(T_A \Psi)^\dagger D_N v_A],$$

$$\mathbb{E} \sum_i K_{A,i}^2 = 6\, \mathbb{E}[v_A^\dagger D_N v_A].$$

These formulas include the spatial factor three and the real-part contraction factor two. The primary calculation checks the first identity against its closed trace formula at every cutoff.

5 Fierz reduction and leading counterterm

Set $s = |\Psi_1|^2 + |\Psi_2|^2$. The Pauli/Fierz identities give

$$\sum_A m_A^2 = s^2, \qquad \sum_A |T_A \Psi|^2 = 3s,$$

$$\sum_A \mathrm{Re}(T_A \Psi)^\dagger v_A = 3s - \frac{s^2}{\rho + \varepsilon_\rho},$$

$$\sum_A |v_A|^2 = 3s - \frac{s^2(\rho + 2\varepsilon_\rho)}{(\rho + \varepsilon_\rho)^2}.$$

For

$$Q_{II} = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

the leading conditional Class-II contraction is

$$\mathbb{E}[e_{\text{II},N} \mid \Psi] = \delta_{\text{cube}} N W_\varepsilon(\Psi) + o(N),$$

$$W_\varepsilon = 9(a + 2b + c)s - \frac{6bs^2}{\rho + \varepsilon_\rho} - \frac{3cs^2(\rho + 2\varepsilon_\rho)}{(\rho + \varepsilon_\rho)^2}.$$

At the production point,

$$a = 0.004499999999998875, \quad b = 0.0018749999999995311, \quad c = 0.002343749999999414,$$

and the eigenvalues of Q_{II} are 0.0012590115009260615 and 0.0055847384990722227. Hence the contraction is nonnegative and nonzero on an open set, so its Gaussian expectation is strictly positive. It follows that

$$\mathbb{E}F_{\mathrm{II},N}/N \longrightarrow \kappa_{II} > 0.$$

The bare positive energy is therefore not uniformly bounded in Gaussian L^1. A vacuum-energy constant cannot cancel the field-dependent local contraction $\delta_{\mathrm{cube}}NW_\varepsilon(\Psi)$. Subtracting this term is the first counterterm candidate for a nondegenerate full-component limit at fixed low-order parameters. It is not proved necessary for every possible weak limit: the bare laws could instead concentrate on the zero set of W_ε. The rational orientation terms do not belong to the current family-mass plus ρ^2, ρ^3 parameter family.

6 Numerical audit and interpretation

The primary audit uses FFT convolution for exact cube multiplicities, eigenspace covariance sums, and tensor Gauss–Hermite quadrature. The non-importing audit uses direct three-dimensional enumeration, original-basis matrix inversion, a different one-dimensional cube integral, and fixed-seed Monte Carlo. They pass 19/19 and 12/12; the integrated audit passes 44/44.

At cutoff $N = 128$, the eigenvalues of D_N/N are between 0.0163072859843 and 0.0163866362110. The high-cutoff full-production diagnostic is

$$\mathbb{E}F_{\mathrm{II},N}/(VN) \simeq 5.42394795319 \times 10^{-4}.$$

The independent dyadic increment is $5.44664650759 \times 10^{-4}$. This decimal coefficient is not an interval evaluation of the C_∞ expectation; only its positivity, formula, and linear exponent are claimed analytically.

7 Renormalisation and composite gates

For graphs with V_D two-derivative vertices, V_0 local vertices, and I internal lines, the superficial degree is

$$\omega = 3L + 2V_D - 4I = 3 - I - V_D - 3V_0.$$

This makes the one-vertex differentiated contraction the leading primitive candidate and leaves open the possibility that one prescribed rational counterterm closes the fixed-floor theory. It does not prove that closure.

There is also a definition problem before squaring K_A. Typical fields have regularity below one half, and

$$K_A = (\rho + \varepsilon_\rho)\nabla\left(m_A/(\rho + \varepsilon_\rho)\right)$$

is a marginal product of the form $C^\alpha C^{\alpha-1}$ with $\alpha < 1/2$. Cutoff convergence and scheme independence require a separate composite lift. The two open gates are therefore 'A6-CLASSII-COUNTERTERM-CLOSURE' and 'A6-CLASSII-K-COMPOSITE-DEFINITION'.

8 Devil's-advocate

1. A factor-two error in complex covariance would change every coefficient. The convention is explicit and hash-audited.

2. A finite-cutoff slope fit cannot prove the theorem. Here the exponent and cube coefficient come from the symbol expansion and Riemann-sum limit; the cutoff table is only a check.
3. A positive density floor does not cure derivative variance. It controls the rational denominator only.
4. Divergent Gaussian mean energy does not prove nonexistence of a renormalised Gibbs measure, nor does it exclude a bare degenerate limit concentrated on $W_\varepsilon = 0$. No such no-go is claimed.
5. Subtracting a constant is inadequate because W_ε is nonconstant and contains new rational terms.
6. The J_A derivative is canonical as a distributional derivative, while K_A remains a marginal rough product. Their continuum statuses are not identified.
7. Family/lock masses change C_∞ but not the high-momentum exponent or universal cube factor.
8. The derivative Class-II energy is not the legacy solver's guarded- quotient diagnostic; the terminology is explicitly separated.

9 Result footer

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Result ID:               A6-CLASSII-UV-POWER-COUNTING
Precise statement:       The canonical bare derivative Class-II Gaussian
                         expectation has a strictly positive linear cutoff
                         divergence with explicit cube coefficient and a
                         nonconstant local leading contraction.
Scope:                   Fixed L=16 torus; full A1 quadratic symbol; three
                         complex fields; six-real Gaussian convention; fixed
                         positive floors; eta_shell=0; sharp cube cutoff.
Dependencies:            A1-KERNEL-IDENTITY;
                         A1-PRODUCTION-FUNCTIONAL-REALISATION.
Evidence grade:          ANALYTIC, EXACT, EXECUTED.
Reproduction command:    python codes/foundations/
                         a6_classii_uv_power_counting_verify.py
Expected output:         19/19 + 12/12; integrated 44/44 PASS.
Falsification gate:      factor/convention error, nonpositive slope, failed
                         asymptotic or independent mismatch, or source drift.
Tier before / after:     new / T4 STRONG-EVIDENCE@BARE-GAUSSIAN-UV.
No-overclaim statement:  No renormalised composite/measure, counterterm
                         necessity for every weak limit, bare-limit
                         classification, sufficiency, regulariser removal,
                         phase transition, BCC, or T5/T6/T7 result.
Next required action:    Operator reproduction; define K_A and prove fixed-
                         floor rational-counterterm stability and tightness.
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